Engine Model(s): PT6A

77-20-00

ENGINE INDICATING - DESCRIPTION AND OPERATION

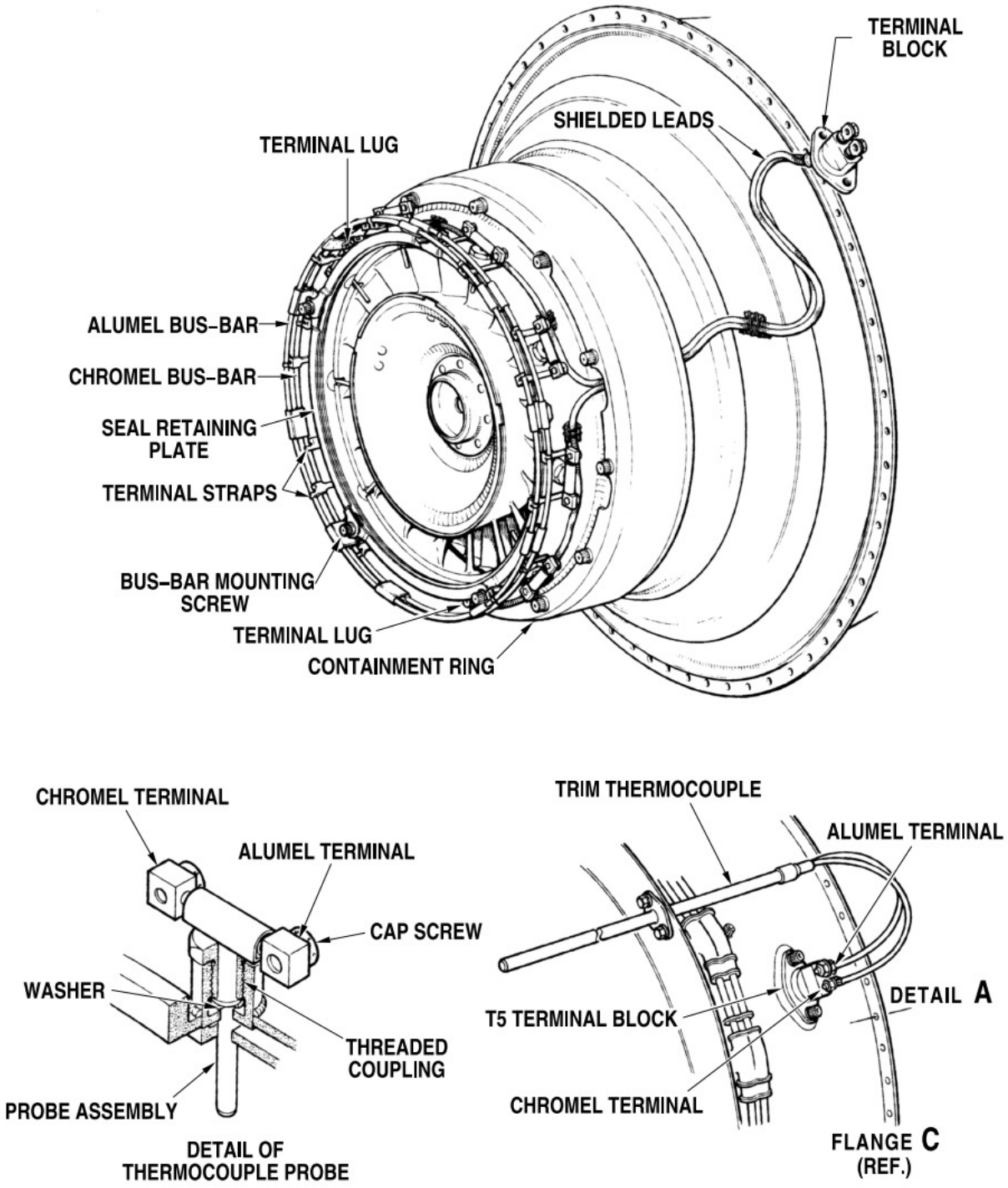
1. General

The interturbine temperature (T5) sensing system provides the pilot with an indication of the engine operating temperature occurring in the zone between the compressor turbine and the power turbine stator. The system consists of a bus-bar assembly, eight individual thermocouple probes connected in parallel, a wiring harness incorporating a terminal block, and a trim harness incorporating a T1 probe. The T1 thermocouple is connected in parallel with the T5 wiring harness to bias the interturbine temperature (ITT) signal, such that the indicated ITT bears a fixed relationship with the compressor inlet temperature and has been derived from measured exhaust temperature and other parameters. This calculated temperature is the true representative of the critical component temperatures in the engine hot section; that is, the biased indicated temperature relates to the critical component temperatures and the biasing T1 probe compensates for sampling error in ITT indication.

2. Description and Operation (Ref. Figs. 1 and 2)

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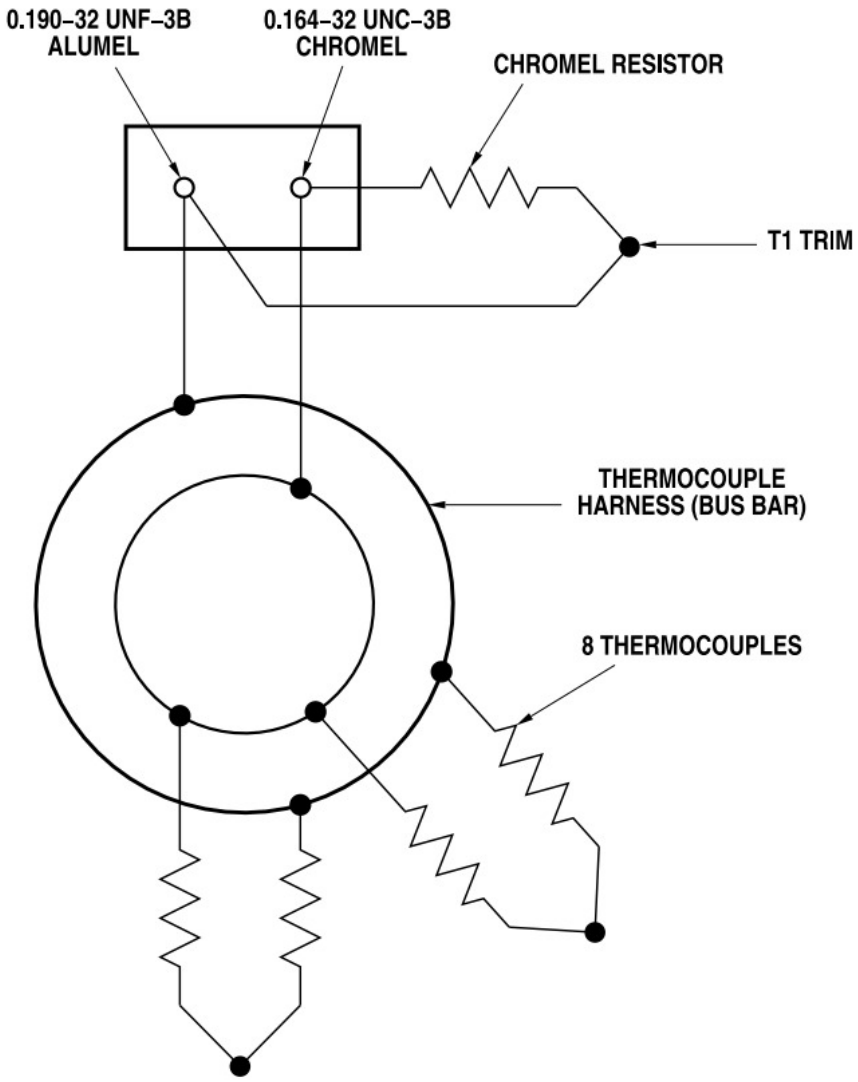
Figure 1 T5 Interturbine Temperature Sensing System



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Figure 2 T5 Schematic Wiring Diagram



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The thermocouple probes consist of eight alumel/chromel assemblies connected in parallel by a bus-bar arrangement. The probes protrude through individual threaded bosses in the power turbine stator housing and into the compressor turbine discharge airstream. Probe locations are at the 1, 2, 3, 4, 8, 9, 10 and 11 o'clock positions on the housing. The probes are secured to their respective bosses by a floating, threaded fitting which forms an integral part of the assembly, each probe being installed in conjunction with a copper gasket.

The bus-bar (Ref. Chap 77-20-01) consists of alumel and chromel rings, isolated and bracketed together to form an assembly; eight straps of respective material are welded to each ring for connection to respective alumel and chromel terminals of the thermocouple probes. Terminal points at the 10 and 5 o'clock positions are provided for connection of the T5 wiring harness. To ensure correct connection of bus-bar straps to probes, alumel terminals of probe are identifiable as having larger screws than the chromel terminals. Likewise, the terminal point at the 10 o'clock position on the alumel ring of the bus-bar is threaded to a larger diameter than the one at the 5 o'clock position on the chromel ring. Also, alumel is slightly magnetic.

The T5 wiring harness (Chap. 77-20-01) consists of a terminal block with two braze-connected leads, the long lead being alumel and the short lead chromel. The terminal lug on the alumel lead has a larger diameter hole than the lug on the chromel lead. The individual leads of the harness are secured to clips welded to the stator housing. The leads are routed forward and secured together to similar clips on the exhaust duct. Prior to mating engine sections at flange C, the wiring harness is routed to the 2 o'clock position within the gas generator case, at which location the terminal block is inserted into a boss and bolted from outside.

The T1 trim thermocouple (Chap. 77-20-01) consists of alumel/chromel leads encased in a stainless steel tube; the chromel lead having an in-line resistor. The required resistance value is determined during final engine acceptance checks when the proper harness resistance is selected. The unit may only be replaced in the field with a trim harness of the same resistance. The trim harness is mounted on the right-hand side of the engine; a mounting flange on the probe casing are connected to their respective terminals on the T5 terminal block.

3. Fault Isolation

For detailed troubleshooting of the engine indicating system, refer to Chapter 72-00-00, FAULT ISOLATION.

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